

Practical Design of an Upper Prosthetic Limb Using Three Dimensional Printer with an Artificial Intelligence Based Controller

A.M.Elbreki
Libyan Authority for Scientific
Research
Benghazi, Libya
nasirburki2013@gmail.com

Safa Ramdan
Dept. of Electrical and Electronic
Engineering (Independent College
Medical Engineering
Benghazi, Libya
safabozer@gmail.com

Faisal Mohamed
Libyan Authority for Scientific
Research
Benghazi, Libya
elabdl@hotmail.com

Khadija Alshari²
Electrical and Electronic Engineering
Department, University of Benghazi
Benghazi, Libya
khadija.imdord@uob.edu.ly

Zakariya Rajab
Electrical and Electronic Engineering
Department, University of Benghazi
Benghazi, Libya
zakreia.hassan@uo.edu.ly

B.Elhub
Faculty of engineering Elmergib
university khoms, Libya
bashirelhub@gmail.com

Abstract—A noticeable rate of people suffer from a disability due to war and other reasons. One group of disabled people is those who have lost their arms, which makes it challenging to perform basic tasks. This paper proposes a solution that will make life easier for amputees. The study is based on the EMG (Electromyography) signal acquired from muscles and the movement detected using HMI (Human Machine Interfacing). The practical arm is designed with a computer-aided design using SolidWorks software. A 3D printer FDM (Fused Deposition Modeling) with Polylactic Acid “PLA” was used to fabricate the Prosthetic Limb using an open-source prototype named Hackberry hand from Exiii Company. An actual circuit design is arranged using simple tools that cost more minor and consists of simple sensors, motors, and Arduino Nano. The findings show that the database extracted 416×4 is presproced classified and tested using KNN (K nearest neighbor) with 94% rate, and obtained 95% with SVM method to recognize four different motions raising the index finger, holding a card with the thumb and forefinger, fist and spread.

Keywords—3D printer, EMG, Prosthetic Arm, SOLIDWORKS, algorithms SVM (Support Vector Machine), KNN (K Neighbors Classifier).

I. INTRODUCTION

Today we live in a world where millions of people have lost limbs due to various factors such as wars, workplace injuries, or diseases such as diabetes, tumor diseases, and others. The percentage of amputees in the country has increased due to armed clashes since the outbreak of the crisis. Where statistics have recorded that nearly 2000 people lose their limbs annually, preventing them from performing their daily jobs [1]. Compared to developing countries that suffer from a massive increase in wars, as is the case in Bangladesh from 2009 to 2016 about 2000 accidents. In 2017, about 7397 people died and were injured. In 2018, deaths increased by 7% over the previous year [2-3]. In response to this reality, technological development has imposed a rush to develop more practical and capable solutions to simulate the sound parties.

A 3D printing provided the designer and manufacturer to customize a model to the patient's specifications [4-5]. The prosthetic limbs (prosthesis) design remains a challenge for the better. Some prosthetics were performed using an intelligent controller for the upper part of the prosthesis based on EMG. The Support Vector Machine (SVM) algorithm

distinguishes various instructions given to the prosthetic arm. SVM seems to perform better than traditional neural network algorithms by implementing the proper training. This paper proposes to design an upper prosthetic limb using artificial intelligence based on 3D printed technology. It's being able to design this prototype and mimic it as a standard function like catching, raising a finger, and relaxing. These movements are controlled using a microcomputer, and the results depend on different algorithms. Sometimes, it's challenging to extract useful features from the remaining muscles of the embolus. More difficulties were added during the multi-layered classification problem. Several researchers suggested and developed different kinds of classification of EMG signals [6]. The electromyography signals from the healthy musculature of the body were wont to determine the orders for this purpose. The EMG signal was extracted for every movement. The EMG signal is classified by a feature-based mapping method. Several pattern recognition techniques are available for functional differentiation, and little research was conducted using multichannel EMG signals apart from the recent work of Kim et. [7]. It has been found that the EMG signals from an increasing number of channels will increase significantly for efficient classification. Some studies [8,9] showed that using multiple channels is beneficial and found that the accuracy of classification on average will increase with increasing channels. A decrease in the return is observed after the number of channels is over four [10]. In a previous study, a computer-aided prosthetic hand model was developed in SolidWorks® that was 3D printed with Poly Lactic Acid (PLA), and the proposed design was closely based on the tendon process associated with human hand function. It can be passed through the classification process to control hand movements, and the filtered data is sent and saved to the Arduino Uno® controller. Then, the Support Vector Machine (SVM) was evaluated as a classifier where the classification accuracy obtained was 96.7%. Electromyography-based grip control was implemented by successfully testing a prosthetic hand designed for three different movements [11-14]. This paper proposes to design an upper limb prosthesis using artificial intelligence, based on 3D printing technology, to design and simulate this prototype as a normal function such as picking up, finger lifting, and relaxation. On the time frequency, they are MAV e (moving average), RMS (root mean square), and WL (waveform length).

II. MATERIALS AND METHODS (AI TOOLS AND ALGORITHMS)

A. SVM support vector machine algorithm

It is a machine learning algorithm for analyzing data for statistical classification. This algorithm is widely utilized in biology as it typically leads to very precise models. In addition, it is more straightforward compared to neural networks. Two steps are required to derive a reliable model using SVM. First a training data set is used to train the algorithm. The training data set contains states; the state is composed of two pieces, one is called the class, and the other is called the attribute. The second step is implemented to test the accuracy of the trained model [12-18]. The goal of SVM is to produce a model based on training data to assign a class to instances using only descriptors. SVM classifiers have four basic concepts:

- Hyperplane:

After obtaining a training data set of two classes, represented as a ray in binary space as shown in Fig.1. The green and red dots represent the first and second entries, respectively. These rays can be separated by a dividing line separating them. In this case, a new state class of an unknown class can be predicted by knowing its location from this space (both sides of the dividing line). Use the name of this super-surface separator line. SVM's job is to find this surface. (called a surface) because it is a surface in the two largest dimensions.

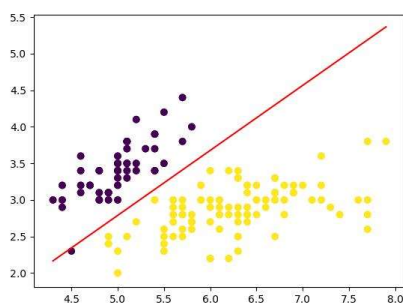


Fig. 1. Dividing lines to separate data in a binary space representation

- Excessive margin – maximum

It is noted that an infinite number of super planes can be drawn. The SVM algorithm chooses the surface in the middle to get the best result in the classification process. The surface with the largest margin between it and the nearest ray in this space is selection in Figure (2,3).

- Flex Margin

In most cases, the rays are not in space so that a superficial surface can separate them, so the SVM algorithm was modified to solve this problem by adding what is known as a soft margin, allowing some rays to be placed in the wrong place without affecting the result.

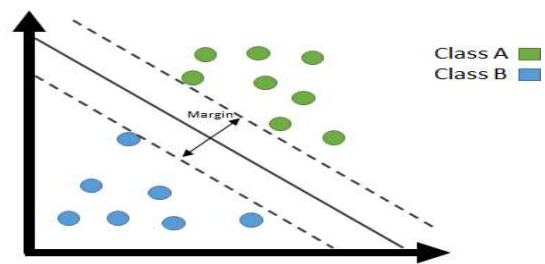


Fig. 2. Selected surface containing the largest margin between it and the nearest ray in space.

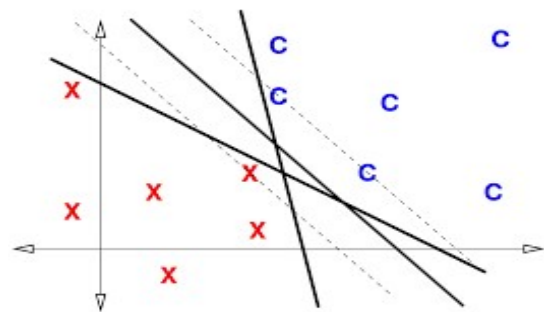


Fig. 3. It represents how to separate data with infinite numbers of superplanes

- Kernel Function

In some cases, the rays cannot be separated by a superficial surface, as shown in Figure 4, but this problem can be solved. Squaring these values and adding the squared values as new dimensions. In this new space, the points are placed so that the rays can be separated by a superficial surface that follows the nucleus that performs the process of projecting the ray space onto an area of larger dimensions so that a superficial surface can separate the X-rays. The squaring operation is the simplest type of kernel function, but it is not used because it doubles the dimensions of space and makes the complexity of the classification process very large. The degree of complexity increases exponentially with the increase in the dimensions of the space. Kernel functions generally offer a middle ground between performance and accuracy in the classification process.

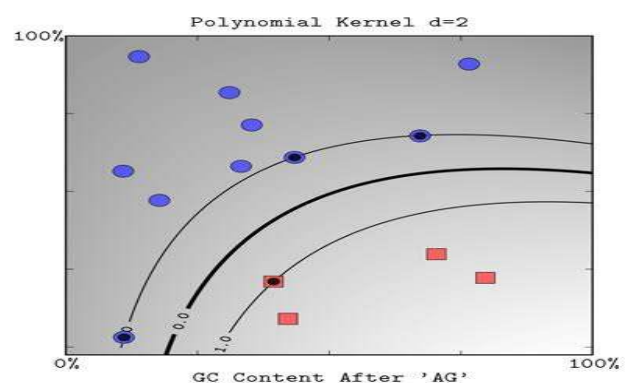


Fig. 4. The projection of an excessive surface into a high-dimensional space

- Nearest Neighbor Algorithm - K

The K-Nearest Neighbor method is a predictive technique suitable for classification models, where K represents the

number of similar states or the number of items in the set. When a new state is introduced to the model, the algorithm searches all the data to find a subset of the most similar states and uses them to predict the output. There are two main determinants of the nearest neighbor algorithm: the number of jars to use (K) and the metric unit of similarity. Each use of the nearest neighbor algorithm requires that a true positive value of K be specified, and this specifies the number of current states that are searched when a new state is presented to the algorithm.

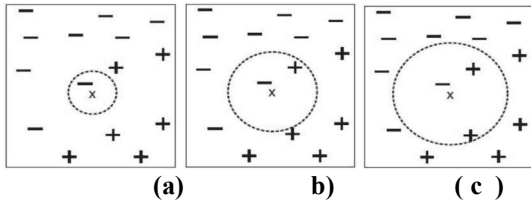


Fig. 5. The neighbor, two neighbors, and the three neighbors closest to the data point. a) neighbor closest to the data point 5, b) The two closest neighbors to the data point 5, c) The three nearest neighbors to the data point.

B. FDM Technology (Extrusion Material)

Figure 6 and Figure 7 show the 3D printing mechanism in the Fused Dimation Modeling (FDM), model. The FDM technology is the most widely used in prosthetic printing worldwide, where the printer is supplied with printing material in the form of wires or filaments attached to a pointed end with a nozzle. This head is heated to melt the printing material, and when moving horizontally and vertically, the material comes out. According to the desired shape, . One of the most famous technologies based on this experience is FDM, and one of the most common materials used in this technology is PLA (multi-block thermoplastics) which is a type of environmentally friendly plastic. ABS (Acrylonitrile Butadiene Styrene) is a kind of environmentally friendly plastic.

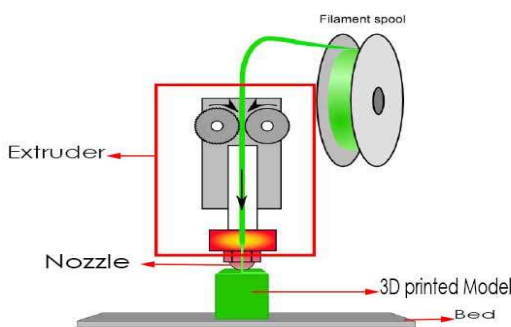


Fig. 6. FDM 3D Printer

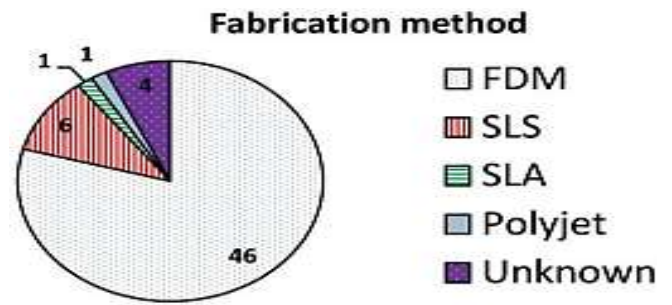


Fig. 7. a statistic of the most commonly used techniques in edge printing

C. Gravity: Analog EMG Sensor by OYMotion

We used the motion EMG sensor shown in Figure 8 to acquire and record muscle signals developed and sold by Thalamic Lab. To achieve the function, the sensor recognized four movements through its classification system: handgrip, the finger spread, index finger raised, and thumb grip with the forefinger.

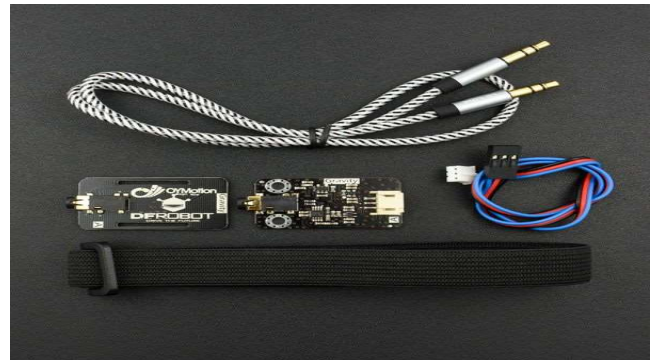


Fig. 8. Oy motion EMG sensor

D. Processing and control system

There are two main processes in the device operating system. The first process is to extract the characteristics, train the classifier and predict the movement. The second process is the control process, which includes moving the motors, and we used the Arduino Nano because it has the analog PWM ports and its small size that is suitable to be placed inside the edge Figure 9 looked at the user.

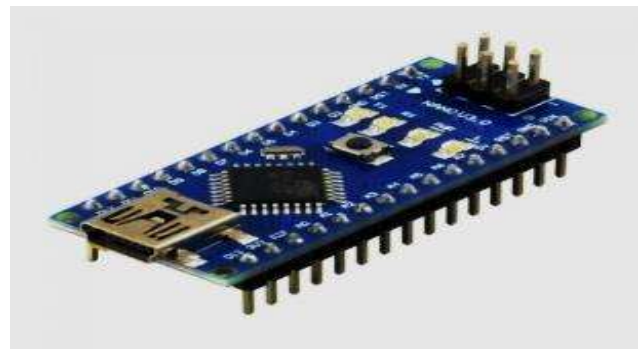


Fig. 9. Arduino Nano

E. Servo Motor

Emax and mini servomotors were used in this study to drive the tip, as shown in Figures 10,a, and 10,b, which are normal DC motors with an integrated mechanical gearbox. It has three

poles; The first is fed by 5 or 6 volts, the second is the servo reference voltage of the motor with a width of PWM, and the last electrode is GND.



Fig 10.a miniServo MG995

Fig 10.b Emax Servo

Fig. 10. A miniServo and Emax Servo

III. RESULT AND DISCUSSION

Hackberry Hand was designed by Japanese startup Exiii, using 3D printing technology. The tip consisted of several mechanical parts and 61 pieces connected by axes and screws. The pieces were printed with 100% filling material for small pieces to make them more durable and withstand forces, and torque was applied with a density of 50% for large pieces to reduce weight Figure 10 After printing some handpieces and setting the print parameters for each design individually using Cura software, this file represents the coordinates that the printer should track and print parameters like print head temperature. Sheet construction, material flow rate, and printing speed.



Fig. 11. Some of the pieces are hand-printed

After printing all the necessary pieces, the fingers and the shirt with the wrist were assembled in Figure 12.



Fig. 12. The parts of the tip after assembly

Three motors were placed in the palm as dictated by design, using an Emax thumb motor and another Emax motor to move the pinky, ring, and middle fingers together, and a motor mini servo to move the index finger. Figure 13 shows the motors placed in the palm after being assembled. The obtained results show the accuracy of each classifier and its uncertainty matrix which represents the accuracy for each individual movement. The diameter of the matrix represents the accuracy of each movement, and the remaining elements in the matrix represent the percentage of uncertainty between

each movement. After training the classifiers mentioned in the Work section, the displayed movements were encoded on axes.

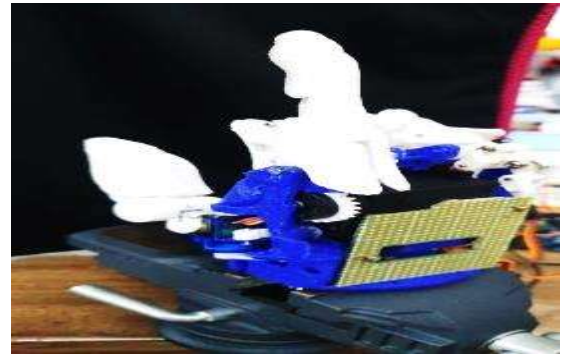


Fig. 13. The actuators are placed inside the palm

After extracting the signal from the sensor and processing it by the Arduino, which then performs the command of the servo motor movements. The data reading appears on the computer screen from which the data is extracted and saved as an excel file with the extension csv (comma separated values), data for each movement. Each movement is recorded separately, and there are a total of four movements. The movements are depicted in Figure 14. The database matrix collected from the patient had dimensions (416×4) after obtaining muscle signal data for the selected movements and saving them in an excel file. Several features are calculated and picked later as mathematical equations: root mean square and wavelength (RMS), wavelength (WL), and mean absolute value (MAV).



Fig. 14. Quoted movements

A. SVM Classifier

The accuracy is 95%, and each movement is displayed in Figures(15, 16, and 17) and table 1.

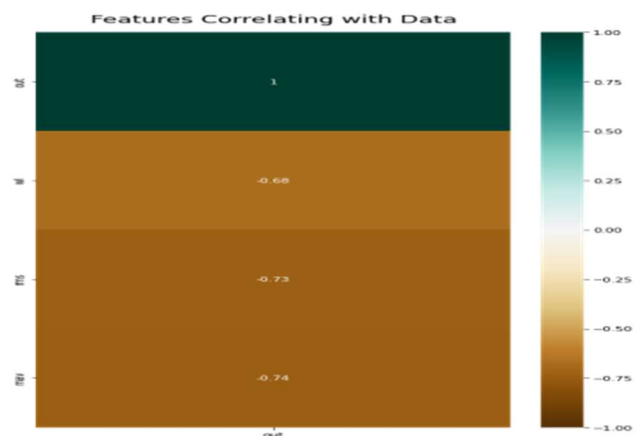


Fig. 15. Relationship between features correlating with data

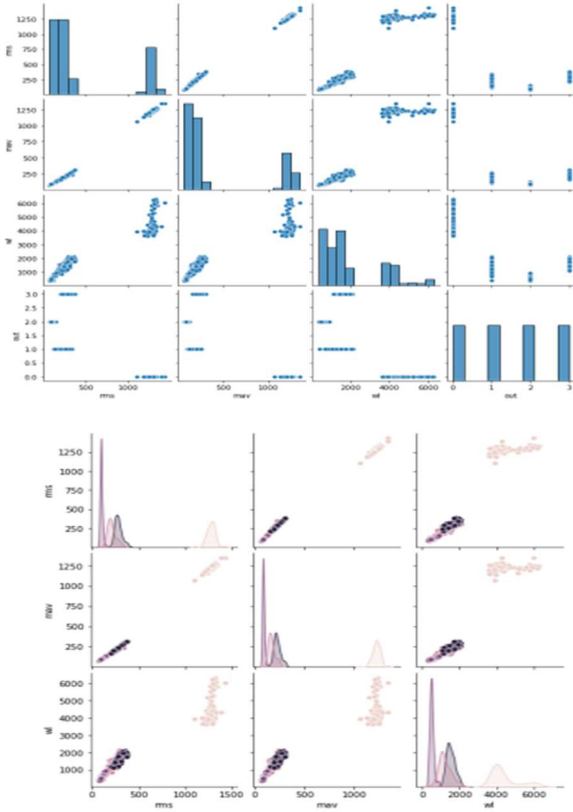


Fig. 16. The figure shows the relationship between features and features and between output.

Table 1 Uncertainty matrix of the SVM classifier after it has been trained

0	1.00	1.00	1.00	36
1	0.89	0.89	0.89	28
2	1.00	1.00	1.00	29
3	0.91	0.91	0.91	32
Accuracy	0.95	125		
macro avg	0.95	0.95	0.95	125
weighted avg	0.95	0.95	0.95	125
Confusion matrix, without normalization	[[36 0 0 0] [0 25 0 3] [0 0 29 0] [0 3 0 29]]			

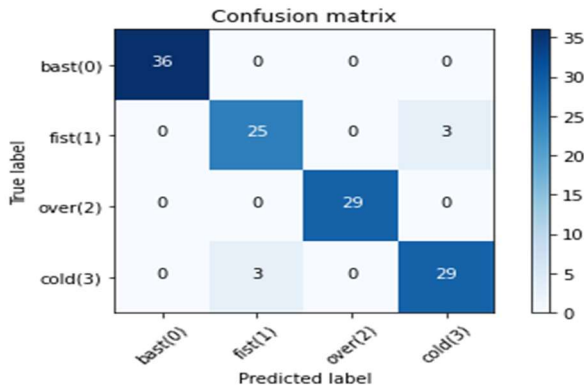


Fig. 17. Uncertainty matrix of the SVM classifier after it has been trained

B. KNN classifier:

accuracy is 94%. The accuracy of each movement was shown in Figure 18 and table 2..

Table 2 Uncertainty matrix of the KNN classifier after it has been trained

0	1.00	1.00	1.00	36
1	0.88	0.82	0.96	28
2	1.00	0.93	1.00	29
3	0.86	0.97	0.91	32
Accuracy	0.94	125		
macro avg	0.95	0.93	0.93	125
weighted avg	0.94	0.94	0.94	125
Confusion matrix, without normalization	[36 0 0 0] [0 23 0 5] [0 0 27 0] [0 1 0 31]]			

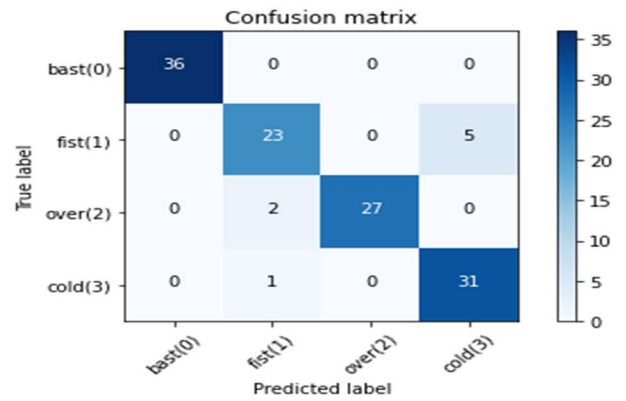


Fig. 18. Uncertainty matrix of the KNN classifier after it has been trained

IV. CONCLUSION

In an age where artificial intelligence techniques, robotic systems, the development of citation methods and vital signs and their greater understanding, along with the emergence of advanced manufacturing techniques, are within the reach of individuals and are no longer confined to large companies and factories, the results obtained from this research work show that today is not Far away when we have a fully functional prosthetic hand. The user must bring the Myoelectric hand close to the object. In this work, an integrated upper prosthesis for a sub-elbow amputation was built that performs the required function, muscle signals were successfully and accurately extracted, and different patterns of muscle signals were interpreted depending on the user-selected movements using SVM and KNN classifiers with accuracy up to 95% and 94% in a row. In general, the tip's response was characterized by speed, accuracy in an acceptable design and a reasonable size for use.

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